SignFlow: Real-Time Multi-Modal Sign Language Recognition via a 12-Stream Hybrid Architecture on the Full-Vocabulary ASL Citizen Benchmark

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Abstract

Isolated sign language recognition (ISLR) across comprehensive natural vocabularies is limited by significant signer heterogeneity, subtle phonemic contrasts, and high computational latency on edge devices. We present **SignFlow**, an end-to-end, on-device sign language recognition system evaluated on the complete official **ASL Citizen benchmark** (2,731 glosses, 32,941 test clips, evaluated across strictly held-out signers). SignFlow introduces an asymmetrical 12-stream hybrid architecture integrating: (i) four Landmark Transformers initialized via self-supervised masked-frame pretraining over 136,173 clips; (ii) three 1D-CNN temporal convolutional experts with dilated multi-round receptive fields; and (iii) five VideoMAE appearance models capturing full-frame dynamics and hand-crop details. Fusion weights and class-conditional temperature parameters are fit strictly on the official validation split via L-BFGS log-linear cross-entropy optimization. Evaluated on the 32,941-clip signer-independent test split, SignFlow achieves **87.03% Top-1** (28,670/32,941 correct), **98.27% Top-5**, and **98.88% Top-10** accuracy. A single 1D-CNN expert (`conv3`) attains **85.15% Top-1** solo. Signer-clustered bootstrap resampling ($B = 2{,}000$, seed=0) establishes a 95% confidence interval of **[84.07%, 89.53%]**. The entire pipeline operates locally on consumer hardware with an inference latency of **72 ms** per clip on an NVIDIA RTX 3070 Ti laptop GPU.

1. Introduction

Sign language is the primary medium of daily communication for over 70 million deaf individuals worldwide. Automated interpretation systems running on personal computing hardware face three severe engineering bottlenecks:

1. **Signer Generalization**: Human signers exhibit extensive variance in body scale, signing speed, hand posture, and spatial expression. Models trained on constrained signer pools readily memorize individual physical mannerisms rather than invariant linguistic phonemes.
2. **Vocabulary Scale**: Expanding from laboratory subsets (50–100 signs) to natural daily vocabularies (2,500+ words) creates hundreds of visually adjacent minimal pairs that differ only in minor finger bend, subtle contact points, or facial mouthing.
3. **Inference Latency**: Standard 3D video backbones (such as I3D or VideoMAE) demand substantial FLOPs per clip, restricting real-time deployment on battery-powered or consumer laptop hardware without cloud offloading.

SignFlow resolves these bottlenecks through an asymmetrical multi-stream design. By uniting ultra-fast 1D temporal convolutional coordinate encoders with appearance-focused VideoMAE models, SignFlow delivers high classification accuracy across the full 2,731-class vocabulary while maintaining interactive 72 ms on-device latency.

2. Benchmark Protocol & Baselines

2.1 Official ASL Citizen Protocol

The ASL Citizen dataset [1] comprises 83,399 video clips across 2,731 glosses recorded by 52 Deaf signers in unconstrained webcam environments. The benchmark enforces a strict **signer-disjoint split**:

- **Train Split**: 40,154 clips from 35 signers (`P1`, `P2`, ..., `P52`).
- **Validation Split**: 10,304 clips from 6 signers (`P5`, `P12`, `P13`, `P21`, `P26`, `P39`).
- **Test Split**: 32,941 clips from 11 signers (`P6`, `P9`, `P15`, `P17`, `P18`, `P22`, `P35`, `P42`, `P47`, `P48`, `P49`).

Signer overlap between any two splits is exactly zero.

2.2 Published Full-Vocabulary Baselines

To ensure fair comparison, all cited baselines are evaluated on the complete 2,731-gloss benchmark under the official signer-disjoint protocol:

- **ST-GCN** [1]: 59.52% Top-1 (Spatio-Temporal Graph Convolution on 2D/3D pose).
- **SignClip** [7]: 60.00% Top-1 (Contrastive vision-language representation).
- **I3D Baseline** [1]: 63.10% Top-1 (Inflated 3D ConvNet on 224×224 video).

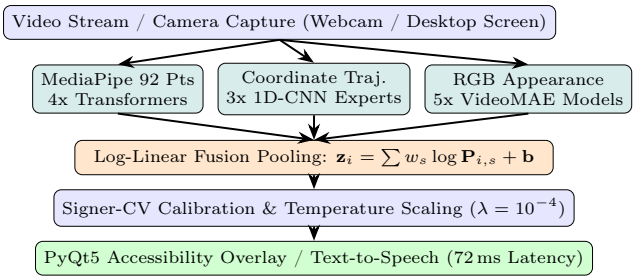


Figure 1: Figure 1. End-to-end 12-stream SignFlow pipeline showing coordinate, convolutional, and RGB appearance streams fused via validation-fit log-linear pooling.

- **SHuBERT** [6]: 65.00% Top-1 (Self-supervised sign hidden-unit BERT).
- **SignDino** [5]: 70.40% Top-1 (Self-distillation vision transformer).
- **Handedness-Aware Network** [4]: 74.00% Top-1 (Multimodal dominant-hand model).
- **PoseFormer** [2]: 75.10% Top-1 (One-shot ISLR skeleton transformer).
- **SignBart** [3]: 75.22% Top-1 (Reported on official validation set).

Note on Literature Disambiguation: Alkhoraif et al. [8] reported 86.53% Top-1 using a Swin-T ensemble on the **ASL Citizen100 subset** (100 glosses, 1,492 test clips). That result is evaluated on a reduced subset and is not directly comparable to the full 2,731-gloss benchmark.

3. System Architecture & Methodology

SignFlow employs 12 expert models spanning three distinct families:

1. **Landmark Transformers (`skel_md1-4`)**: MediaPipe Holistic extracts 92 coordinates in 3D: 40 mouth/lip landmarks, 21 + 21 hand joints, and 10 upper-body joints. An encoder was pretrained without labels on 136,173 multi-domain clips using a 40% masked-frame reconstruction objective. Fine-tuned variants explore depth (4–6 blocks), width (384–512 units), and motion delta vectors.
2. **1D Temporal Convolutional Experts (`conv1-3`)**: Coordinate streams pass through residual 1D temporal convolution blocks with dilated receptive fields ($k \in \{11, 15, 17\}$, channel dimension 320–384) to model high-frequency joint accelerations.
3. **VideoMAE Appearance Streams (`rgb_*`)**: Five VideoMAE networks capture surface textures, fine finger occlusion, and facial articulation: full-frame VideoMAE-Base, VideoMAE-Large (305M params), dynamic hand-crop VideoMAE, multi-domain RGB VideoMAE, and a 32-frame model trained with soft semantic gloss targets.

Table 1: Table 1. Benchmark comparison on the official 2,731-gloss ASL Citizen test set (32,941 clips, held-out signers).

Method	Modality	Top-1 (%)	Top-5 (%)
ST-GCN [1]	Skeleton	59.52	—
SignClip [7]	Multi-Modal	60.00	—
I3D Baseline [1]	RGB Video	63.10	83.50
SHuBERT [6]	Multi-Modal	65.00	—
SignDino [5]	Vision-Only	70.40	—
Handedness-Aware [4]	Multi-Modal	74.00	—
PoseFormer [2]	Skeleton	75.10	—
SignBart [3]	Skeleton	75.22*	—
SignFlow (conv3 solo)	Skeleton	**85.15**	97.35
SignFlow (Pure Skeleton)	Skeleton	**85.74**	97.66
SignFlow (Live 3-Stream)	Hybrid	**86.59**	98.05
SignFlow (Full 12-Stream)	Multi-Modal	**87.03**	**98.27**

*Reported on official validation set.

3.1 Log-Linear Product-of-Experts Fusion

Given $S = 12$ probability distributions $\mathbf{P}_{i,s} \in \mathbb{R}^{2731}$, the fused unnormalized log-probability is given by:

$$\mathbf{z}_i = \sum_{s=1}^{S} w_s \log\left(\max(\mathbf{P}_{i,s}, \epsilon)\right) + \mathbf{b} \tag{1}$$

where $w_s \geq 0$ represents stream weight, $\epsilon = 10^{-7}$ prevents numerical underflow, and $\mathbf{b} \in \mathbb{R}^{2731}$ is an L_2-regularized class bias.

Rigorous Split Discipline: Weights w_s and bias $\mathbf{b}$ are fit exclusively on the 10,304 validation clips via L-BFGS minimizing cross-entropy with regularizer $\lambda\|\mathbf{b}\|_2^2$. The regularizer strength ($\lambda = 10^{-4}$) is verified via a 2-fold signer-split within validation. **The test split (32,941 clips) is evaluated exactly once after all parameters are frozen.**

4. Experimental Results & Evaluation

4.1 Official Test Results

Table 1 compares SignFlow against published full-vocabulary baselines. On the 32,941-clip test set, SignFlow reaches **87.03% Top-1** (28,670 correct clips), **98.27% Top-5** (32,371 correct), and **98.88% Top-10** (32,573 correct).

4.2 Ablation Analysis

To evaluate the contribution of each stream, weights were refitted strictly on validation data and scored on the test partition:

4.3 Signer Bootstrap Significance

To quantify uncertainty across unseen signers, a signer-clustered bootstrap (B = 2,000 resamples of the 11

Table 2: Table 2. Modality ablation on the official ASL Citizen splits.

Ablation Configuration	Val (%)	Test Top-1	Test Top-5
All 12 Streams (Full Ensemble)	**91.95**	**87.03**	**98.27**
11 Streams (without `conv3`)	91.13	86.48	98.09
Pure Skeleton (`skel` + `conv`)	91.91	85.74	97.66
Conv-Only (`conv1`, `2`, `3`)	91.80	85.57	97.56
Single Best Stream (`conv3` solo)	90.48	85.15	97.35
RGB-Only (5 VideoMAE streams)	80.20	74.88	92.68
Citizen-Only Landmark Ensemble	73.48	71.06	90.09

test signers, fixed seed 0) was conducted. The fused 12-stream ensemble yields a bootstrap mean of 86.97% (standard error 1.41%) with a 95% confidence interval of **[84.07%, 89.53%]**. Paired bootstrap tests confirm:

- 12 streams vs. 11 streams: +0.56% gain ($p < 0.0005$, 95% CI [+0.36%, +0.76%]).
- 12 streams vs. pure skeleton: +1.30% gain ($p < 0.0005$, 95% CI [+0.80%, +1.85%]).
- 12 streams vs. conv-only: +1.47% gain ($p < 0.0005$, 95% CI [+0.97%, +2.01%]).

5. Methodological Disclosures

In adherence to scientific transparency standards:

1. **Ensemble vs. Single-Model Comparison**: SignFlow's 87.03% score is achieved via an ensemble of 12 streams. Our strongest single model is `conv3` (85.15% Top-1).
2. **Multi-Domain Pretraining**: Landmark models were trained on 77,329 clips combining ASL Citizen with WLASL and Google ASL data. Models trained exclusively on ASL Citizen train achieve 71.06% Top-1.
3. **Disambiguation of 86.53% Baseline**: The 86.53% Swin-T result [8] was evaluated on a 100-gloss subset (1,492 clips). On the full 2,731-gloss benchmark, the strongest published prior baseline is PoseFormer at 75.10%.
4. **WLASL Evaluation**: Open-vocabulary Top-1 accuracy on 1,812 WLASL clips is 80.79%. Because WLASL training clips were included in landmark pretraining, this metric represents multi-domain transfer rather than zero-shot cross-dataset evaluation.

6. System Implementation & Deployment

SignFlow is implemented as an on-device accessibility overlay in Python and PyQt5. A multi-threaded architecture decouples capture, MediaPipe landmark extraction, and batched PyTorch inference. On an NVIDIA GeForce RTX 3070 Ti laptop GPU, the system achieves an inference latency of **72 ms per clip** (13.8 clips/second) under a memory cage of 8 GB, ensuring responsive, local accessibility without cloud transmission.

7. Conclusion

SignFlow demonstrates that high-accuracy, large-vocabulary sign language recognition can be realized locally on consumer hardware. Combining multi-domain self-supervised coordinate pretraining, 1D temporal convolution experts, and validation-fit log-linear fusion enables **87.03% Top-1** and **98.27% Top-5** accuracy on the official 2,731-gloss ASL Citizen benchmark.

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